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End-of-Module Exam: 2027 Spring Semester

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Number of supplementary answer sheets attached:

Total number of added pages:

Module:	w.BA.XX08-WIN.XX / Data Science Introduction
Exam date / time:	Practice set
Duration:	approx. 120 Minutes (practice scope)
Total number of pages:	2
Permissible resources:	Closed book
Calculator:	no calculator
Dictionary:	Yes

General information:

- Use a black or blue pen (invisible ink is not allowed) to write your answers.
- All options of a question (i.e. A–D) can be ticked.
- ONLY tick ONE option per question in the table.
- Scoring: each correctly answered question = 1 point.
- Write your answers only on the answer sheet provided; rough work on question pages is not scored.
- Switch off your mobile phone and put it in your bag for the duration of the examination.
- If you want to write a draft, use the back of the question pages. Only the answer sheet counts.
- Please remain seated until all exams have been collected.

Marked by:	Total points:	Grade:

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Grading Table

Performance Assessment	Weighting	Grade	Weighted Grade
Performance Assessment 1	100%		
		<i>(Grade on cover sheet)</i>	
Module grade (Evento)			⇐ Rounded ⇐

Marking Table

No.	Task	Points	Corrector's, initials
1	MC written Exam	90 Points	hitz
	Total	90 Points	

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Answer sheet

ONLY the answer sheet will be scored. All options of a question (i.e. A–D) can be ticked.

ONLY tick ONE option per question in this table.

Scoring: Each correctly answered question is worth **1 point**.

Example: For question 0, C should be ticked, then it would look like this.

Question	Answer	Question	Answer
1.	(A) (B) (C) (D)	46.	(A) (B) (C) (D)
2.	(A) (B) (C) (D)	47.	(A) (B) (C) (D)
3.	(A) (B) (C) (D)	48.	(A) (B) (C) (D)
4.	(A) (B) (C) (D)	49.	(A) (B) (C) (D)
5.	(A) (B) (C) (D)	50.	(A) (B) (C) (D)
6.	(A) (B) (C) (D)	51.	(A) (B) (C) (D)
7.	(A) (B) (C) (D)	52.	(A) (B) (C) (D)
8.	(A) (B) (C) (D)	53.	(A) (B) (C) (D)
9.	(A) (B) (C) (D)	54.	(A) (B) (C) (D)
10.	(A) (B) (C) (D)	55.	(A) (B) (C) (D)
11.	(A) (B) (C) (D)	56.	(A) (B) (C) (D)
12.	(A) (B) (C) (D)	57.	(A) (B) (C) (D)
13.	(A) (B) (C) (D)	58.	(A) (B) (C) (D)
14.	(A) (B) (C) (D)	59.	(A) (B) (C) (D)
15.	(A) (B) (C) (D)	60.	(A) (B) (C) (D)
16.	(A) (B) (C) (D)	61.	(A) (B) (C) (D)
17.	(A) (B) (C) (D)	62.	(A) (B) (C) (D)
18.	(A) (B) (C) (D)	63.	(A) (B) (C) (D)
19.	(A) (B) (C) (D)	64.	(A) (B) (C) (D)
20.	(A) (B) (C) (D)	65.	(A) (B) (C) (D)
21.	(A) (B) (C) (D)	66.	(A) (B) (C) (D)
22.	(A) (B) (C) (D)	67.	(A) (B) (C) (D)

23.	(A) (B) (C) (D)	68.	(A) (B) (C) (D)
24.	(A) (B) (C) (D)	69.	(A) (B) (C) (D)
25.	(A) (B) (C) (D)	70.	(A) (B) (C) (D)
26.	(A) (B) (C) (D)	71.	(A) (B) (C) (D)
27.	(A) (B) (C) (D)	72.	(A) (B) (C) (D)
28.	(A) (B) (C) (D)	73.	(A) (B) (C) (D)
29.	(A) (B) (C) (D)	74.	(A) (B) (C) (D)
30.	(A) (B) (C) (D)	75.	(A) (B) (C) (D)
31.	(A) (B) (C) (D)	76.	(A) (B) (C) (D)
32.	(A) (B) (C) (D)	77.	(A) (B) (C) (D)
33.	(A) (B) (C) (D)	78.	(A) (B) (C) (D)
34.	(A) (B) (C) (D)	79.	(A) (B) (C) (D)
35.	(A) (B) (C) (D)	80.	(A) (B) (C) (D)
36.	(A) (B) (C) (D)	81.	(A) (B) (C) (D)
37.	(A) (B) (C) (D)	82.	(A) (B) (C) (D)
38.	(A) (B) (C) (D)	83.	(A) (B) (C) (D)
39.	(A) (B) (C) (D)	84.	(A) (B) (C) (D)
40.	(A) (B) (C) (D)	85.	(A) (B) (C) (D)
41.	(A) (B) (C) (D)	86.	(A) (B) (C) (D)
42.	(A) (B) (C) (D)	87.	(A) (B) (C) (D)
43.	(A) (B) (C) (D)	88.	(A) (B) (C) (D)
44.	(A) (B) (C) (D)	89.	(A) (B) (C) (D)
45.	(A) (B) (C) (D)	90.	(A) (B) (C) (D)

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MC Data Science Introduction (90 Points)

1. Which of the following best distinguishes the Viable System Model (VSM) from System Dynamics (SD)? (1)
 - A. VSM simulates behaviour over time, while SD is essentially a static organisational chart.
 - B. VSM is a recursive management structure for viability, while SD is a simulation method for behaviour over time.
 - C. Neither VSM nor SD makes any use of feedback.
 - D. They are two names for exactly the same method.
2. Which statement about VSM and System Dynamics is NOT correct? (1)
 - A. The VSM is a recursive structure for viable systems.
 - B. Both approaches ignore feedback entirely.
 - C. They are complementary systems-thinking approaches.
 - D. System Dynamics models system behaviour over time.
3. Which option correctly pairs each approach with its primary purpose? (1)
 - A. VSM -> simulate behaviour over time; System Dynamics -> ensure organisational viability.
 - B. VSM -> ensure organisational viability through recursive control; System Dynamics -> simulate dynamic behaviour over time.
 - C. Both are methods for storing data for reporting.
 - D. VSM -> clustering; System Dynamics -> classification.
4. In the light of Ashby's Law, what does 'adaptation' mean? (1)
 - A. Collecting more data without changing capacity.
 - B. Increasing one's own variety (or attenuating the environment's) so the system matches environmental variety.
 - C. Reducing the system to a single fixed state.
 - D. Ignoring environmental change.
5. Which statement about Ashby adaptation is NOT correct? (1)
 - A. A system can amplify its own variety in order to adapt.
 - B. Adaptation aligns internal variety with environmental variety.
 - C. Failure to adapt can let the environment dominate the system.
 - D. The only valid response is to reduce one's own variety.

6. An organisation can adapt to a more variable environment by... (1)
- A. removing all of its feedback channels.
 - B. increasing its own variety or reducing/filtering the environment's variety.
 - C. guaranteeing its survival without changing anything.
 - D. ignoring the environment and freezing all of its processes.
7. Which statement correctly distinguishes Big Data and Data Science? (1)
- A. Big Data fully contains Data Science.
 - B. Data Science is only the storage layer of Big Data.
 - C. They are synonyms.
 - D. Big Data refers to data characteristics and infrastructure, while Data Science is a broader, transdisciplinary discipline.
8. Which statement is NOT correct regarding the Data Science distinction? (1)
- A. Big Data and Data Science mean exactly the same thing.
 - B. Data Science is a broad, transdisciplinary discipline.
 - C. Analytics is one component within Data Science.
 - D. Big Data refers primarily to data traits and infrastructure.
9. Which statement about the relationship between Big Data and Data Science is NOT correct? (1)
- A. Big Data mainly concerns data volume, variety, velocity and veracity.
 - B. Data Science is a broader, transdisciplinary discipline.
 - C. Big Data and Data Science are interchangeable terms.
 - D. Analytics is one component within Data Science.
10. Which of the following best describes a real-time adaptive model? (1)
- A. A spreadsheet pivot table.
 - B. A model trained once and never updated again.
 - C. A model that continuously adapts to streaming data via feedback.
 - D. A static end-of-month report.
11. Which statement about real-time adaptive models is NOT correct? (1)
- A. They rely on continuous feedback loops.
 - B. They are well suited to fast-changing (VUCA) environments.
 - C. Once deployed they never need updating because they are 'adaptive' by name only.
 - D. They can incorporate reinforcement learning.
12. A fraud-detection system that updates continuously as new transactions stream in best illustrates... (1)
- A. a static, once-trained model.
 - B. online / real-time adaptation to streaming data.
 - C. purely descriptive reporting of the past.
 - D. manual rule-based control with no data.

13. A model scores 98% on the training set but only 70% on the test set. This most likely indicates... (1)
- A. overfitting / high variance.
 - B. perfect generalization.
 - C. underfitting.
 - D. high bias.
14. Which statement about overfitting is NOT correct? (1)
- A. It reflects high variance.
 - B. It generalizes poorly to new data.
 - C. It is best fixed by making the model even more complex.
 - D. Cross-validation helps detect it.
15. Which of the following is the best remedy for a model that is clearly overfitting? (1)
- A. Increase the model's complexity even further.
 - B. Remove the test set so the gap disappears.
 - C. Train and test the model on the same data.
 - D. Add regularization, gather more data, or simplify the model.
16. Which statement about visualizing results is correct? (1)
- A. A well-designed chart can never mislead.
 - B. A truncated axis can exaggerate small differences, violating the true-and-fair view.
 - C. Visualizations always inform objectively.
 - D. Visual literacy is never required to read a chart.
17. Which statement about visualization is NOT correct? (1)
- A. Dashboards help reduce data overload.
 - B. Visualizations are always intuitive and need no visual literacy.
 - C. Visualizations can be biased by design choices.
 - D. A misleading axis violates objective presentation.
18. Which statement about responsible data visualization is NOT correct? (1)
- A. A professional-looking chart is automatically objective and free of bias.
 - B. A truncated y-axis can mislead even when the numbers are correct.
 - C. The 'true and fair view' requires fair presentation, not only accuracy.
 - D. Visual literacy affects how a chart is interpreted.
19. Which of the following is a hallmark of complex CPS data? (1)
- A. Small, perfectly structured, independent samples.
 - B. Static data with no time dependency.
 - C. Large-scale, high-frequency, heterogeneous and non-IID data.
 - D. A single clean spreadsheet.

20. Which statement about complexity is NOT correct? (1)
- A. CPS data is often non-IID (dependent across time and context).
 - B. Complexity affects the value and interpretability of data.
 - C. Complex systems have many interdependent variables.
 - D. Complexity is the same as a complicated machine with many parts.
21. Which example best illustrates 'non-IID' data? (1)
- A. Sensor readings whose values depend on previous time steps and context.
 - B. A balanced two-class dataset.
 - C. A series of independent coin flips.
 - D. A perfectly shuffled, independent random sample.
22. What does class weighting do? (1)
- A. Duplicates minority-class samples to enlarge the class.
 - B. Interpolates synthetic minority samples.
 - C. Deletes majority-class examples.
 - D. Increases the minority class's weight in the loss function without changing the data.
23. Which statement about class weighting is NOT correct? (1)
- A. It is useful for imbalanced problems such as fraud detection.
 - B. It physically duplicates or deletes data rows.
 - C. It helps prevent the majority class from dominating.
 - D. It modifies the training loss.
24. How does class weighting differ from oversampling? (1)
- A. Class weighting deletes majority data, while oversampling reweights the loss.
 - B. Both physically enlarge the dataset with extra rows.
 - C. They are exactly the same technique.
 - D. Class weighting reweights the loss function, while oversampling adds (duplicated) data.
25. A team says 'we have lots of data - what can we build?'. A target-oriented coach would reframe this as... (1)
- A. keep exploring the data until something emerges.
 - B. choose the most complex algorithm available.
 - C. skip stakeholder alignment to save time.
 - D. start from the goal and design data, features and metrics around it.
26. Which statement about target-oriented action is NOT correct? (1)
- A. The cycle culminates in a concrete action.
 - B. Measurable targets help track progress.
 - C. The goal drives data and method choices.
 - D. Projects succeed best by starting from available data and ignoring the goal.

27. Which approach is NOT consistent with target-oriented (goal-first) working? (1)
- A. Ending the analysis cycle with a concrete action.
 - B. Collecting whatever data is available first and hoping a goal emerges.
 - C. Aligning stakeholders on a shared target.
 - D. Defining a measurable outcome before choosing the data.
28. Explainable AI (XAI) primarily aims to... (1)
- A. hide the model's internal logic.
 - B. explain why a model made a decision, supporting user trust.
 - C. maximize predictive accuracy only.
 - D. remove human oversight.
29. Which statement about explainability is NOT correct? (1)
- A. XAI explains the 'why' behind decisions.
 - B. Explanations should adapt to the audience.
 - C. Opaque black-box models are ideal for high-stakes use.
 - D. It is key to trust and transparency.
30. Why does explainability matter most for high-stakes AI decisions? (1)
- A. It mainly speeds up model training.
 - B. It lets users understand and challenge the 'why', building trust and accountability.
 - C. It removes the need for any human oversight.
 - D. It guarantees the model is always correct.
31. In control terms, Ashby's Law requires that... (1)
- A. the environment be ignored.
 - B. the controller be simpler than the system it steers.
 - C. the controller have at least as much variety as the system it steers.
 - D. more data alone guarantees control.
32. Which statement about Ashby control is NOT correct? (1)
- A. The controller's variety must match the system's.
 - B. Cybernetics underpins steering through feedback.
 - C. CEP and SPC are cyber-layer control functions.
 - D. A low-variety controller reliably manages a high-variety system.
33. A controller can steer a system effectively only if... (1)
- A. it is simpler than the system it controls.
 - B. it ignores feedback from the system.
 - C. its variety is at least as large as the system's variety.
 - D. it holds more data than the system, regardless of variety.

34. The Viable System Model (VSM) is best described as... (1)
- A. a confusion matrix.
 - B. a one-off linear plan.
 - C. a star schema for data warehousing.
 - D. a recursive management structure that ensures viability via anti-oscillatory feedback.
35. Which statement about the VSM is NOT correct? (1)
- A. It uses anti-oscillatory feedback.
 - B. It is recursive across levels.
 - C. It is identical to a static organisational chart with no feedback.
 - D. It supports adaptability and learning.
36. Which statement about the Viable System Model (VSM) is NOT correct? (1)
- A. It is a one-off linear plan that excludes feedback.
 - B. It maps well onto multi-level cyber-physical systems.
 - C. Anti-oscillatory feedback supports stability and learning.
 - D. It is recursive across organisational levels.
37. Class imbalance is dangerous mainly because... (1)
- A. it removes all features from the data.
 - B. it makes every model too simple.
 - C. a majority-only predictor can show high accuracy while ignoring the minority class.
 - D. accuracy becomes impossible to compute.
38. Which statement about class imbalance is NOT correct? (1)
- A. SMOTE interpolates new minority samples.
 - B. High accuracy always means the minority class is well predicted.
 - C. The minority class is often the important one.
 - D. Resampling and class weighting can help.
39. A rare-disease model (2% positive) reports 98% accuracy but never flags a sick patient. The core issue is... (1)
- A. using too many clusters.
 - B. class imbalance, which makes accuracy misleading (very poor recall).
 - C. underfitting caused by too few features.
 - D. data leakage from the test set.
40. In the VUCA framework, 'uncertainty' means... (1)
- A. unclear meanings with no precedents.
 - B. future outcomes are unknown despite available data.
 - C. rapid, unstable change.
 - D. many interrelated variables.

41. Which statement about uncertainty is NOT correct? (1)
- A. Outcomes are unknown despite available data.
 - B. Scenario analysis helps cope with it.
 - C. Uncertainty implies analysis is fundamentally impossible.
 - D. It is distinct from ambiguity.
42. Which situation best matches 'uncertainty' in the VUCA sense? (1)
- A. Unclear meaning with no precedents at all.
 - B. Many tightly coupled variables interacting at once.
 - C. Sudden, fast and unstable change.
 - D. Unknown future outcomes although relevant data already exists.
43. Unsupervised learning... (1)
- A. finds hidden structure or groupings in unlabeled data.
 - B. always learns from rewards and penalties.
 - C. predicts known labels for new data.
 - D. requires labeled data.
44. Which statement about unsupervised learning is NOT correct? (1)
- A. It requires labels for every example.
 - B. Clustering is a common application.
 - C. It is harder to evaluate than supervised learning.
 - D. It works on unlabeled data.
45. A retailer groups customers by behaviour with no predefined labels. This is... (1)
- A. linear regression.
 - B. supervised classification.
 - C. reinforcement learning.
 - D. unsupervised clustering.
46. Ashby's Law of Requisite Variety states that... (1)
- A. simpler systems are always better.
 - B. environmental complexity is irrelevant to survival.
 - C. only variety can absorb variety - a controller must match the system's variety.
 - D. data volume alone guarantees control.
47. Which statement about requisite variety is NOT correct? (1)
- A. More data automatically equals requisite variety.
 - B. The controller must match the system's variety.
 - C. Too little variety leads to being overwhelmed by the environment.
 - D. Variety is a measure of complexity (distinguishable states).

48. What happens when the environment's variety exceeds a system's variety? (1)
- A. The system automatically adapts and thrives.
 - B. The environment dominates and can ultimately overwhelm the system.
 - C. Nothing changes for the system.
 - D. The system's larger data volume restores control.
49. System Dynamics studies... (1)
- A. static snapshots only.
 - B. labeled classification problems.
 - C. how systems behave over time via stocks, flows, feedback and delays.
 - D. database table schemas.
50. Which statement about System Dynamics is NOT correct? (1)
- A. It uses stocks, flows and feedback loops.
 - B. Vensim/STELLA are typical modeling tools.
 - C. It is a static, time-free reporting method.
 - D. It models behaviour over time.
51. Which pairing is correct in System Dynamics? (1)
- A. Stock = accumulation; Flow = rate of change.
 - B. Delays have no effect on system behaviour.
 - C. Balancing loop = amplifies change; Reinforcing loop = dampens change.
 - D. Stock = rate of change; Flow = accumulation.
52. A smart factory is best described as... (1)
- A. a manual workshop with no data.
 - B. a CPS-based, connected production enabling real-time monitoring and control (Industry 4.0).
 - C. a single spreadsheet.
 - D. an offline database of past orders.
53. Which statement about the smart factory is NOT correct? (1)
- A. It is a purely offline reporting system.
 - B. It integrates physical and cyber layers.
 - C. MES bridge machines and ERP systems.
 - D. Predictive maintenance reduces unplanned downtime.
54. Which statement about a smart factory is NOT correct? (1)
- A. It is essentially an offline reporting database.
 - B. It enables real-time monitoring and optimization of production.
 - C. MES connect physical machines with cyber ERP systems.
 - D. Predictive maintenance helps avoid unplanned downtime.

55. On a strongly imbalanced dataset, which single metric is most misleading on its own? (1)
- A. Accuracy
 - B. Recall
 - C. F1 score
 - D. Precision
56. Which statement about model evaluation is NOT correct? (1)
- A. The confusion matrix underlies many classification metrics.
 - B. Recall measures the proportion of predicted positives that are correct.
 - C. The ROC curve shows the trade-off between hit rate and false alarms.
 - D. Recall matters most when false negatives are costly.
57. For a screening test where missing a positive case is very costly, which metric matters most? (1)
- A. Training accuracy.
 - B. Precision.
 - C. Recall (sensitivity).
 - D. The number of input features.
58. Which of the following is an unsupervised clustering algorithm? (1)
- A. K-means
 - B. Logistic regression
 - C. Linear regression
 - D. A decision-tree classifier
59. Which statement about unsupervised algorithms is NOT correct? (1)
- A. Fuzzy C-means forces each point into exactly one cluster.
 - B. For K-means, k is chosen beforehand.
 - C. K-means uses centroids as cluster centres.
 - D. K-medoids is more robust to outliers than K-means.
60. Which statement describes K-medoids correctly? (1)
- A. It always outperforms every other clustering method.
 - B. It uses the cluster mean as the centre.
 - C. It requires labeled data to run.
 - D. It uses an actual data point as the cluster centre, which makes it robust to outliers.
61. The target-centric approach is best described as... (1)
- A. a data-first exploration with no defined goal.
 - B. a linear, one-shot analysis.
 - C. a structured, iterative method that continuously refines insights as new data emerges.
 - D. a pure visualization tool.

62. Which statement about the target-centric approach is NOT correct? (1)
- A. It is iterative and refines insights over time.
 - B. It recommends starting from data and letting goals emerge later.
 - C. It aligns activities and data with goals.
 - D. It centres on customer value and clear priorities.
63. Which of the following is NOT a feature of the target-centric approach? (1)
- A. Iterative refinement of insights as new data arrives.
 - B. Allocating resources according to strategy-derived priorities.
 - C. Organising systems around business subjects (e.g., customer or product).
 - D. Starting from whatever data is available with no defined goal.
64. Which best distinguishes clustering from network analysis? (1)
- A. Clustering groups similar items, while network analysis studies the connections between items.
 - B. Both require fully labeled data.
 - C. Clustering needs labels and network analysis is supervised.
 - D. They are identical techniques.
65. Which statement about clustering and network analysis is NOT correct? (1)
- A. Clustering groups similar data points.
 - B. Communities in graphs resemble clusters.
 - C. Both require labeled data to function.
 - D. Network analysis focuses on connections and structure.
66. Which option best distinguishes clustering from network analysis? (1)
- A. Both require fully labeled data.
 - B. They are two names for the same technique.
 - C. Clustering groups similar items, while network analysis studies relationships and connections.
 - D. Clustering studies edges, while network analysis forms centroids.
67. Which of the following is a Big Data myth (i.e., NOT true)? (1)
- A. About 95% of big data is unstructured.
 - B. 'More data is always better.'
 - C. Veracity is an essential dimension of Big Data.
 - D. Quality and relevance matter, not just volume.
68. Which statement about Big Data hype is NOT correct? (1)
- A. Much of big data is unstructured.
 - B. Data lakes need metadata to remain usable.
 - C. Bigger data is always better regardless of quality.
 - D. Volume alone does not guarantee value.

69. Which of the following is a TRUE statement (not a myth) about Big Data? (1)
- A. Most big data is unstructured, and data quality (veracity) matters.
 - B. Merging data sources automatically produces a single, complete 'Golden Record'.
 - C. A data lake is inherently valuable even without metadata.
 - D. More data is always better, regardless of quality.
70. Exploratory Data Analysis (EDA) is mainly about... (1)
- A. deploying the final model.
 - B. writing the business case.
 - C. tuning hyperparameters only.
 - D. understanding the data (distributions, missing values, outliers) before modeling.
71. Which statement about EDA is NOT correct? (1)
- A. It should be skipped to save time on large datasets.
 - B. It examines distributions and outliers.
 - C. EDA helps spot data-quality issues early.
 - D. It informs cleaning and feature engineering.
72. Which statement about Exploratory Data Analysis (EDA) is NOT correct? (1)
- A. It examines distributions, missing values and outliers.
 - B. It precedes the modeling phase.
 - C. It guides later feature engineering.
 - D. It may freely use the test set during training to improve features.
73. Which of the following is a common MISINTERPRETATION of Ashby's Law? (1)
- A. It is a guiding principle of cybernetics.
 - B. Attenuating the environment's variety is a valid response.
 - C. You must always simply add more complexity to your own system.
 - D. The controller must match the system's variety.
74. Which statement about Ashby's Law is a misinterpretation (i.e., NOT correct)? (1)
- A. Variety is simply the volume of data you hold.
 - B. You can attenuate environmental variety instead of amplifying your own.
 - C. The controller should match the system's variety.
 - D. Variety measures the number of distinguishable states.
75. Which is a CORRECT understanding of Ashby's Law (i.e., avoids the common misinterpretation)? (1)
- A. You can match variety by amplifying your own variety OR by attenuating the environment's.
 - B. You must always keep adding internal complexity to your own system.
 - C. The law is an exact numeric formula to be solved.
 - D. Variety is simply the same as the volume of data you hold.

76. VUCA stands for... (1)
- A. Variety, Uniqueness, Consistency, Accuracy.
 - B. Volume, Uncertainty, Complexity, Analytics.
 - C. Velocity, Utility, Clarity, Agility.
 - D. Volatility, Uncertainty, Complexity, Ambiguity.
77. Which statement about VUCA is NOT correct? (1)
- A. Volatility means fast, unstable change.
 - B. Data Science helps organisations sense, model and respond.
 - C. Uncertainty means unknown outcomes despite data.
 - D. The four dimensions all mean the same thing.
78. Which assignment of a VUCA dimension to its meaning is correct? (1)
- A. Ambiguity = rapid, unstable change.
 - B. Uncertainty = unclear meaning with no precedents.
 - C. Volatility = unknown outcomes despite available data.
 - D. Complexity = many interrelated variables with non-obvious cause and effect.
79. Which of the following is a lazy learner? (1)
- A. Decision tree
 - B. Neural network
 - C. Logistic regression
 - D. K-nearest neighbors (KNN)
80. Which statement about learner types is NOT correct? (1)
- A. Eager learners generalize during training.
 - B. Eager learners are memory-efficient at inference time.
 - C. KNN is a typical eager learner with very fast predictions.
 - D. Lazy learners store data and generalize at query time.
81. Which statement about eager and lazy learners is NOT correct? (1)
- A. Lazy learners tend to use more memory.
 - B. Eager learners generally predict slowly while lazy learners predict instantly.
 - C. Lazy learners (e.g., KNN) defer the work to prediction time.
 - D. Eager learners build a model during training.
82. Which of the following describes sampling bias? (1)
- A. Using a sample that is not representative of the target population.
 - B. Considering only successful cases.
 - C. Over-trusting automated outputs.
 - D. Emphasizing data that confirms one's beliefs.

83. Which statement about bias is NOT correct? (1)
- A. Bias mitigation includes documenting assumptions.
 - B. AI automation removes all bias.
 - C. Survivorship bias ignores failures.
 - D. Confirmation bias favours data supporting prior beliefs.
84. A credit model trained on historically biased approvals keeps disadvantaging one group. This is primarily... (1)
- A. random statistical noise.
 - B. proof that the model is unbiased.
 - C. overfitting only.
 - D. historic bias that is amplified algorithmically.
85. In the Viable System Model, early warning is provided mainly by... (1)
- A. anti-oscillatory feedback loops that detect and dampen instability early.
 - B. a star schema.
 - C. ignoring feedback.
 - D. deleting historical data.
86. Which statement about VSM early warning is NOT correct? (1)
- A. Early detection enables timely intervention.
 - B. Removing feedback improves early warning.
 - C. SPC and anomaly detection aid early warning.
 - D. Anti-oscillatory loops dampen emerging instability.
87. Which statement about early warning through feedback is NOT correct? (1)
- A. Removing feedback gives the system earlier warnings.
 - B. Anti-oscillatory loops dampen instability early.
 - C. SPC and anomaly detection can flag deviations in time.
 - D. Incomplete system knowledge can cause a control breakdown.
88. Data Science is best described as... (1)
- A. a transdisciplinary field (math/statistics, computer science/IT, domain knowledge, social science).
 - B. only software engineering.
 - C. purely applied statistics.
 - D. only domain knowledge.
89. Which statement about interdisciplinarity is NOT correct? (1)
- A. Data Science draws on multiple disciplines.
 - B. Domain knowledge helps prevent 'solitary analysis'.
 - C. Data Science is purely a branch of software engineering.
 - D. Social science contributes ethics and fairness perspectives.

90. Which option best captures why Data Science is described as transdisciplinary? (1)
- A. Domain knowledge is irrelevant to it.
 - B. It is only applied statistics.
 - C. It is purely a branch of software engineering.
 - D. It integrates math/statistics, computing, domain knowledge and social science to solve complex problems.